Municipal Public Cleaning Expenditure and Solid Waste Management: Evidence from Peru, 2015–2019

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Abstract

This study estimates the effect of municipal public cleaning expenditure on solid waste management across Peruvian districts from 2015 to 2019. The main identification threat is expenditure endogeneity: municipalities with stronger institutional capacity simultaneously execute higher budgets and report better outcomes. To correct this bias, we apply the Correlated Random Effects with Control Function (CRE+CF) estimator proposed by **Wooldridge2015**, combining the **Mundlak1978** device with first-stage residuals as an additional regressor. The analysis covers three management dimensions using a panel of 9,322 observations, 1,874 municipalities, and 196 provincial clusters. A 10% increase in accrued expenditure raises daily waste collection by 8.79% (elasticity = +0.879; $p < 0.001$), increases the municipal planning index by 0.030 points on a five-instrument scale ($\hat{\beta} = +0.300$; $p < 0.001$), and raises the probability of formal sanitary landfill disposal by 2.2 percentage points over an 18% baseline (AME = +0.022; $p < 0.001$). All

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three primary hypotheses survive Bonferroni-Holm correction. Results identify municipal finance as the binding constraint for environmental formalization of final disposal, and support reforms to Peru's FONCOMUN transfer formula targeting municipalities with the largest solid waste management deficits.
Keywords: municipal expenditure, solid waste management, sanitary landfill, CRE+CF estimator, Mundlak device, Peru, fiscal decentralization, FONCOMUN

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1. Empirical Strategy

1.1. Identification and Baseline Model

The main empirical strategy uses a Two-Way Fixed Effects (TWFE) design with Correlated Random Effects (CRE) and a control function (CF) to correct for endogeneity of municipal expenditure (**Wooldridge2015**). The general model for municipality i in year t is:

$$Y_{it} = \alpha + \beta \ln G_{it} + \gamma \bar{G}_i + \delta_t + \varepsilon_{it} \tag{1}$$

where Y_{it} is the solid waste management outcome (FR, QPRS, CSRS, PI, RS, or PRS), $\ln G_{it} = \ln(\text{Expenditure}_{it} + 1)$ is the log of accrued public cleaning expenditure, $\bar{G}_i$ is the time mean of $\ln G_{it}$ for municipality i (the Mundlak device), and δ_t are year fixed effects.

1.2. Endogeneity Correction: CRE + Control Function

Municipal expenditure may be correlated with unobservable factors (institutional capacity, environmental preferences), generating endogeneity. Following **Wooldridge2015**, we implement the correction in two stages:

Stage 1 (pooled OLS with Mundlak means):

$$\ln G_{it} = \pi_0 + \pi_1 \bar{G}_i + \sum_t \pi_t \delta_t + v_{it} \tag{2}$$

The residuals $\hat{v}_{it}$ capture expenditure variation unexplained by permanent municipal heterogeneity.

Stage 2: For nonlinear models (Ordered Probit, Probit, Fractional Logit), $\hat{v}_{it}$ is included as an additional regressor. If $\hat{v}_{it}$ is statistically significant, it confirms endogeneity and justifies the CF correction.

Standard errors are clustered at the provincial level (province-clustered SE) to correct for serial correlation and heteroscedasticity at the municipality level.

1.3. Identification Assumptions

Stable Unit Treatment Value Assumption (SUTVA)

To ensure the validity of the CRE+CF estimates, the SUTVA assumption is maintained. Since solid waste management and budget execution in Peru fall under the strict district-level competence of the Organic Law of Municipalities (Law No. 27972), we assume that expenditure in district i generates no spillover effects on waste management outcomes in district j. This premise is reasonable because: (i) public cleaning services are non-exportable by jurisdictional definition; (ii) final disposal infrastructure (sanitary landfills) operated predominantly at the district scale during 2015–2019; and (iii) waste concession contracts did not cross district boundaries in the analysis period, according to RENAMU records.

Exogeneity of the FONCOMUN Instrument

As a robustness check, the FONCOMUN intergovernmental transfer is used as an instrument for $\ln G_{it}$. The exclusion restriction requires that FONCOMUN affects waste management outcomes only through public cleaning expenditure, not directly. Following **ConleyHansenRossi2012**, we report plausible exogeneity bounds to assess the robustness of the exclusion restriction to local deviations.

Parallel Trends (TWFE)

TWFE validity requires that, in the absence of expenditure variation, municipalities would have followed parallel trajectories in the outcome variables. We verify this assumption through pre-trend tests using the lag $\ln G_{it-1}$ (DL(1) specification) and the event study analysis presented in Appendix section Appendix .

1.4. Models by Dimension

D1 — Collection Efficiency

For collection frequency (FR: 1=daily, 5=never), we estimate an Ordered Probit CRE+CF given the ordinal nature of the variable. For quantity collected (QPRS, in kg/day), we use a log-log OLS model with a zero-collection indicator (corner solution). For service coverage (CSRS: 1<25%, 4>75%), we estimate an Ordered Probit CRE+CF analogous to that for FR.

D2 — Municipal Planning

The planning index (PI: sum of 5 binary instruments, range 0–5) is estimated with OLS CRE+CF. Additionally, each binary component (PIGARS,

PMRS, SRRS, PTRS, PSFRSRS) is modeled with Probit CRE+CF to identify which instruments respond to expenditure.

D3 — Adequate Final Disposal

For sanitary landfill use (RS: binary), we use Probit CRE+CF. For the proportion of waste sent to sanitary landfills (PRS $\in [0, 1]$), we apply a Fractional Logit (**PapkeWooldridge1996**) with the control function integrated.

1.5. Multiple Testing Correction

The three primary hypotheses (H1: effect on FR; H2: effect on PI; H3: effect on PRS) are tested simultaneously using the Bonferroni-Holm correction to control the familywise error rate. Secondary comparisons within each dimension (D2 components, D1 sub-models) are reported without multiplicity correction, following standard practice for pre-registered exploratory secondary analyses.

1.6. Two-Stage Bootstrap Inference

For nonlinear second-stage models (Ordered Probit, Probit, and Fractional Logit), analytical standard errors understate true sampling variability because they do not incorporate the estimation error of $\hat{v}_{it}$ from the first stage (**Wooldridge2015**). To correct this inference bias, we implement a two-stage bootstrap with $B = 999$ replications, resampling by provincial cluster in both stages simultaneously: in each replication, the first stage (OLS with Mundlak means) is re-estimated to obtain a new $\hat{v}_{it}^{(b)}$, which is then incorporated into the corresponding second stage. The standard errors reported in table 1 and in the component tables are the standard deviations

of the bootstrap distribution of the coefficients and average marginal effects (AME). For OLS models (QPRS and PI), analytical province-clustered standard errors are asymptotically valid (**Wooldridge2015**) and are reported directly; the bootstrap confirms concordance with analytical standard errors in both cases.

1.7. Identification Frontier

The identifying variation in the CRE+CF estimator comes from within-municipality expenditure fluctuations, net of the municipality time mean ($\bar{G}_i$): formally, the estimator exploits $\tilde{G}_{it} = \ln G_{it} - \bar{G}_i$, i.e., transitory deviations of expenditure from each municipality's historical level. This component is purged of permanent unobservable heterogeneity — structural institutional capacity, geography, fiscal history — which the Mundlak device absorbs in the first stage.

The identification frontier that the design cannot cross consists of time-varying confounders correlated with $\tilde{G}_{it}$. The archetypal example is a change in local political capacity: a newly elected, technically competent mayor may simultaneously increase public cleaning expenditure and waste management quality in the same period, generating a spurious correlation that the control function cannot eliminate because the residual $\hat{v}_{it}$ does not distinguish between an exogenous expenditure shock and a management capacity shock. Robustness checks address this risk from multiple angles: the DL(1) test rules out pre-existing trends and anticipation effects; the statistical significance of $\hat{v}_{it}$ in all models confirms the direction of the bias; and the **Oster2019** bounds quantify the magnitude of the static omitted variable bias required to nullify the effects (Appendix 4). However, none of these tests definitively

rules out a coincident time-varying confounder. The estimated effects should be interpreted as conditional effects under the CRE assumption — that no heterogeneous unobserved temporal trends persist across municipalities — rather than as local average treatment effects in the strict sense. Integration of the FONCOMUN transfer as an external instrument is the identification extension that would recover the LATE for the subgroup of municipalities whose expenditure responds to variations in that transfer, converting the conditional effects reported here into verifiable local causal effects.

2. Results

2.1. Endogeneity Diagnosis

The first step before interpreting structural coefficients is to verify whether the CRE+CF correction is empirically justified. In all estimated models, the coefficient on $\hat{v}_{it}$ — first-stage residuals — is statistically significant at the 1% level ($p < 0.001$). This result rejects the null hypothesis of expenditure exogeneity and confirms that the uncorrected TWFE estimator produces biased estimates. Without the control function, the correlation between $\ln G_{it}$ and unobservable institutional factors would overestimate the expenditure effect: municipalities with greater management capacity tend to execute higher budgets while simultaneously reporting better waste indicators, so the naive OLS captures both the expenditure effect and the pre-existing institutional advantage. The CRE+CF estimation decomposes these sources of variation and isolates the expenditure effect under the CRE assumption that no heterogeneous unobserved temporal trends persist across municipalities.

First-stage parameters are presented in table .3 of the Appendix. The

pooled OLS with Mundlak means ($\bar{G}_i$) and year fixed effects explains 74% of the variance in $\ln G_{it}$. Mundlak means are highly significant (coeff. = 0.921, $p < 0.001$), indicating that permanent municipal heterogeneity — captured by $\bar{G}_i$ — accounts for the bulk of expenditure variation. The residual variation $\hat{v}_{it}$, representing transitory deviations of expenditure from each municipality's historical level, is the identifying component in the second stage.

2.2. Dimension 1 — Collection Efficiency

Collection Frequency (FR)

Municipal public cleaning expenditure has a statistically significant and economically substantive effect on waste collection frequency. The Ordered Probit CRE+CF coefficient is $\hat{\beta} = -0.522$ ($p < 0.001$), where the negative sign reflects variable coding: $FR = 1$ corresponds to daily collection (maximum frequency) and $FR = 5$ to no collection. Higher expenditure shifts municipalities toward lower values of the ordinal index, i.e., toward higher collection frequencies. The magnitude of the latent coefficient exceeds the uncorrected TWFE estimate by a factor of 6.5 ($\hat{\beta}_{\text{TWFE}} = -0.081$), illustrating the attenuation bias introduced by endogeneity when institutional heterogeneity is not controlled.

The result is consistent with Hypothesis H1 under the CRE assumption: higher municipal public cleaning expenditure increases collection frequency. This effect plausibly operates through the direct operational channel: higher expenditure enables fuel procurement, fleet maintenance, and field staff payroll — all binding constraints for service frequency in budget-constrained municipalities.

Daily Quantity of Waste Collected (QPRS)

The estimated expenditure elasticity for daily quantity of waste collected (QPRS) is +0.879 (SE = 0.021, $p < 0.001$), implying that a 10% increase in municipal expenditure is associated with an 8.79% increase in kilograms of waste collected per day. This result is consistent with the hypothesis of decreasing returns: municipalities with low baseline expenditure exhibit the largest marginal gains in effective coverage. The elasticity is close to but significantly below unity (t-test of $H_0 : \hat{\beta} = 1$, $p < 0.001$), suggesting that operational capacity does not scale perfectly proportionally with budget — presumably due to frictions in hiring skilled personnel and the availability of treatment infrastructure in rural municipalities. Compared to the TWFE estimator ($\hat{\beta}_{\text{TWFE}} = 0.027$), the CRE+CF correction raises the elasticity by more than thirty-fold, demonstrating that attenuation bias is quantitatively dominant for this outcome.

The 33× amplification for QPRS contrasts with the 7–8× ratios observed for PI and RS, and requires an economic explanation. QPRS is a log-scale flow variable whose total variance is dominated by the between-municipality dimension: the difference between a 5,000-resident municipality without its own fleet and a 50,000-resident municipality with daily service generates cross-sectional variation of one to two orders of magnitude, while within-municipality temporal variation is comparatively small. The TWFE estimator absorbs all between-municipality variation into individual fixed effects, identifying only from transitory expenditure deviations — which for QPRS are small and noisy, producing a near-zero TWFE coefficient (+0.027). The CRE+CF also recovers the between dimension through Mundlak means, multiplying

the identifying signal. For PI and RS, temporal variation is relatively larger: PIGARS adoption or the transition to sanitary landfill are discrete events occurring within the panel's observation window and are therefore partially identified by the TWFE, explaining their smaller amplification ratios. In short, the asymmetry in amplification ratios reflects differences in the within-to-between variance ratio across outcomes, not an anomaly in CRE+CF estimation.

Service Coverage (CSRS)

The CRE+CF coefficient on service coverage is $\hat{\beta} = +0.254$ ($p < 0.001$). Since CSRS is coded with increasing values toward higher coverage ($CSRS = 4$ corresponds to coverage above 75% of the district territory), the positive sign indicates that expenditure expands the geographic reach of the service. The magnitude is smaller than the effect on QPRS, consistent with a cost structure in which territorial service expansion (incorporating new collection zones) is marginally more expensive than increasing service intensity in already-covered areas. Results for FR and CSRS are presented in table .6.

2.3. Dimension 2 — Municipal Planning

The solid waste planning index (PI) responds more strongly to expenditure than any other outcome variable in the model. The OLS CRE+CF coefficient is $\hat{\beta} = +0.300$ (SE $= 0.015$, $p < 0.001$), interpretable directly: a 10% increase in accrued public cleaning expenditure increases the planning index by 0.030 points (on a 0 to 5 scale). The TWFE estimator underestimated this effect by nearly eight-fold ($\hat{\beta}_{\text{TWFE}} = 0.039$), confirming that endogeneity artificially compresses coefficients when unobservable institutional heterogeneity is corre-

lated with both expenditure and planning instrument adoption. The result is consistent with Hypothesis H2 under the CRE assumption and suggests that municipal expenditure operates partly through an institutional channel: financial resources facilitate hiring technical staff, conducting diagnostics, and formalizing management procedures that translate into the adoption of mandatory planning instruments.

Component analysis reveals heterogeneity in the response. The PIGARS (Municipal Solid Waste Environmental Management Plan) shows the largest average marginal effect among the five binary instruments: AME = +0.055 ($p < 0.001$), indicating that a 10% expenditure increase raises by 5.5 percentage points the probability that a municipality has an approved PIGARS. This result has a direct policy interpretation: the PIGARS is the most resource-intensive planning instrument (requiring diagnostics, consulting, and citizen participation processes), making financing a binding constraint for its adoption in lower-fiscal-capacity municipalities. Component results are presented in table .7.

2.4. Dimension 3 — Adequate Final Disposal

Sanitary Landfill Use (RS)

The average marginal effect on the probability of sanitary landfill use (AME = +0.022, $p < 0.001$) reveals that financing constitutes a material barrier to the environmental formalization of final disposal. A 10% expenditure increase raises by 2.2 percentage points the probability that a municipality disposes its waste in formal infrastructure, validating H3. In absolute terms, given that the median municipality in the 2015–2019 period generated approximately 5,500 annual tons of solid waste, this marginal increase is equivalent to redirecting

around 110 tons per year from open dumpsites to controlled disposal. The comparison with the TWFE estimator is revealing: the uncorrected coefficient on RS is -0.003 — not only insignificant but with the opposite sign — confirming that reverse causality bias (municipalities with worse disposal indicators receive more corrective resources) completely masked the positive expenditure effect when endogeneity was not controlled.

Proportion of Waste in Sanitary Landfill (PRS)

The Fractional Logit CRE+CF (**PapkeWooldridge1996**) estimates an average marginal effect of $+0.026$ ($p < 0.001$) on the proportion of waste sent to sanitary landfills. This effect complements that of RS: while the AME on RS captures the extensive margin (probability of adopting formal disposal), the AME on PRS captures the intensive margin (utilization rate among municipalities already with access to sanitary landfills). The coexistence of positive and significant effects at both margins indicates that expenditure not only incorporates new municipalities into formal disposal, but also deepens usage among those already practicing it. This result rules out the alternative hypothesis that the RS effect is purely mechanical (municipalities building a new cell automatically report higher PRS): if that were the case, the PRS effect should concentrate among entering municipalities and be zero for already-formalized ones.

2.5. Main Results Table

table 1 consolidates CRE+CF coefficients for the six outcomes across three dimensions. For ease of comparison, uncorrected TWFE coefficients are also reported. The systematic difference between both columns confirms

the bias direction described in section 2.1: TWFE underestimates effects in D1 and D2, and reverses the sign in D3, in all cases due to the positive correlation between observed expenditure and unobserved institutional factors that deteriorate outcomes.

2.6. Parallel Trends Verification

The validity of the CRE+CF estimator in a five-year panel requires that, in the absence of expenditure variation, municipalities would have followed parallel trajectories in the outcome variables. We verify this assumption through the DL(1) distributed lag specification, which adds the lag $\ln G_{it-1}$ to the baseline model and examines whether prior-period expenditure predicts current-period outcomes once contemporaneous expenditure is controlled:

$$Y_{it} = \alpha + \beta_0 \ln G_{it} + \beta_1 \ln G_{it-1} + \gamma \bar{G}_i + \delta_t + \varepsilon_{it} \tag{3}$$

If anticipation exists or if past expenditure affects outcomes through channels distinct from the contemporary one, $\hat{\beta}_1$ should be statistically significant. Results rule this out for all three primary outcomes. For QPRS, the lag coefficient is $\hat{\beta}_1 = +0.012$ (SE $= 0.019$, $p = 0.53$), while the contemporaneous coefficient remains virtually unchanged relative to the baseline model ($\hat{\beta}_0 = +0.871$, $p < 0.001$). For PI, $\hat{\beta}_1 = +0.009$ (SE $= 0.011$, $p = 0.41$), with $\hat{\beta}_0 = +0.294$ ($p < 0.001$). For RS, the AME on the lag is $+0.002$ ($p = 0.61$), versus the contemporaneous AME of $+0.021$ ($p < 0.001$).

In all three cases, the lag coefficient is statistically indistinguishable from zero and quantitatively small relative to the contemporaneous effect (ratios $\hat{\beta}_1/\hat{\beta}_0$ of 0.014, 0.031, and 0.095, respectively). These results are

Table 1: Effect of Public Cleaning Expenditure on Municipal Solid Waste Management
CRE+CF vs. Uncorrected TWFE Models, Peru 2015–2019

	Dimension 1 — Collection			**D2**	**Dimension 3 — Disposal**	
	FR	*QPRS*	*CSRS*	*PI*	*RS*	*PRS*
CRE+CF						
$\hat{\beta}$ / AME	-0.522	$+0.879$	$+0.254$	$+0.300$	$+0.022$	$+0.026$
SE	(clust.)	(0.021)	(clust.)	(0.015)	(clust.)	(clust.)
p-value	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001
TWFE						
$\hat{\beta}$	-0.081	$+0.027$	$+0.116$	$+0.039$	-0.003	-0.003
Observations	$9,322$	$9,322$	$9,322$	$9,322$	$9,225$	$9,322$
Municipalities	$1,874$	$1,874$	$1,874$	$1,874$	$1,874$	$1,874$
Clusters	196	196	196	196	196	196
$\hat{v}_{it}$ sig.	Yes	Yes	Yes	Yes	Yes	Yes

Notes. Treatment variable is $\ln G_{it}$: log of accrued public cleaning expenditure. FR = collection frequency (ordinal 1–5; 1=daily); QPRS = kg of waste collected per day (log-log); CSRS = service coverage (ordinal 1–4); PI = planning index (0–5 scale, OLS); RS = sanitary landfill use (Probit, AME reported); PRS = proportion of waste in landfill (Fractional Logit, AME reported). CRE+CF: Correlated Random Effects with control function (**Wooldridge2015**); Mundlak device included in all models. Standard errors clustered at the provincial level (196 clusters). All models include year fixed effects and controls: ln(population), urbanization rate, provincial poverty rate, ln(own revenues per capita), provincial capital dummy. Source: RENAMU 2015–2019 (INEI) and SIAF/MEF.

consistent with the no-anticipation assumption and support interpreting the contemporaneous effects as causal responses to current-period expenditure, not artifacts of pre-existing trends. The event study analysis with a four-year window is presented in Appendix 4.

2.7. Multiple Testing Correction

The three primary research hypotheses — H1 (effect on FR), H2 (effect on PI), and H3 (effect on PRS) — are tested simultaneously under the Bonferroni-Holm correction to control the familywise error rate. Adjusted p-values for all three hypotheses are $p_{\text{adj}} < 0.001$. The rejection of H1, H2, and H3 is therefore robust to multiple comparisons: the results are not artifacts of multi-outcome analysis. Component analyses within each dimension (PIGARS, PMRS, SRRS, and other binary D2 instruments; CSRS as a D1 sub-outcome) are reported without multiplicity correction, following standard practice for pre-registered secondary exploratory comparisons.

2.8. Plausibility Bounds: Oster (2019)

As a complementary sensitivity analysis, we calculate the selection proportionality bounds δ^* from **Oster2019** for the three primary outcomes (complete results in section Appendix). The δ^* statistic quantifies how much more influential unobservables would need to be than observables to nullify the estimated effect. All three results yield $\delta^* < 0$, with $|\delta^*| > 1.35$ in all cases (QPRS: $\delta^* = -1.75$; PI: $\delta^* = -1.80$; RS: $\delta^* = -1.35$).

The finding $\delta^* < 0$ has a precise interpretation: it validates the robustness of the OLS benchmark (the TWFE $\rightarrow$ CRE+CF transition) against static omitted variable bias. Permanent unobservables would need to operate in the

opposite direction from observables — and be more than 1.35 times more influential — to reverse the coefficient movement between the two specifications. This condition contradicts the bias direction documented by the control function test: municipalities with greater unobservable institutional capacity tend to spend *more*, not less, on public cleaning. The result $\delta^* < 0$ thus provides a solid foundation for the CRE+CF estimation: static unobservable selection, if present, would reinforce the estimated coefficients rather than reduce them to zero. The residual threat — time-varying confounders — is not ruled out by the Oster test and is discussed in section 1.7.

3. Discussion

3.1. Effect Magnitudes and Comparative Benchmarks

The estimated elasticity of +0.879 between municipal expenditure and daily waste collected falls in the upper range of effects documented in the solid waste management cost and efficiency literature. **Bel2010**, in their analysis of Catalan municipalities, document that economies of scale in waste management are exhausted above populations of approximately 20,000 inhabitants, implying decreasing expenditure-output elasticities in larger municipalities. The elasticity of +0.879 estimated for Peru is consistent with that framework: municipalities operating below the minimum efficient scale threshold exhibit near-proportional returns. **Kinnaman2006** argues, from a welfare economics perspective, that underprovision of collection services in low-income municipalities reflects budget constraints rather than preferences; the results of this paper support that reading for the Peruvian case: the expenditure-QPRS elasticity close to unity (+0.879) indicates that collection

capacity is approximately proportional to expenditure, so the coverage gap observed between rich and poor municipalities is primarily attributable to the fiscal gap, not to differences in technical efficiency.

The elasticity of +0.879 falls in the mid-to-high range of the comparative evidence from low- and middle-income countries (LMICs), and its position within that range is interpretable in terms of institutional structure. **Guerrero2013**, in a comparative study of waste management systems in cities across Latin America, Sub-Saharan Africa, and Asia, document that the primary constraint for formal collection in low-fiscal-capacity municipalities is operational financing, not technological availability or the regulatory framework. In the Sub-Saharan contexts analyzed — where fiscal decentralization has advanced in parallel with the transfer of solid waste competencies to local governments, as in Ghana, Kenya, and Tanzania — cities with more stable intergovernmental transfers exhibit collection coverage rates 15 to 30 percentage points higher than comparable cities with volatile financing. Evidence from the *What a Waste 2.0* report (**Kaza2018**) corroborates this pattern for 368 cities in low- and lower-middle-income countries: collection coverage is strongly correlated with municipal per capita operating expenditure, with a log-log slope of 0.7–0.9 for the poorest city quartile — a range that coincides with the Peruvian elasticity of +0.879.

The contrast with South Asian contexts, particularly India, illustrates why the Peruvian elasticity is mid-to-high rather than low. In secondary urban municipalities of Indian states with mixed formal-informal provision, marginal fiscal expenditure faces a substitution effect with the informal sector that depresses the effective elasticity: when the municipality formalizes

collection, it displaces pre-existing informal providers, and the net increase in waste collected is smaller than the additional expenditure would suggest (**Kaza2018**). In rural Peru, that substitution is minimal: the smallest district municipalities have no organized informal collection sector, so each additional sol in expenditure translates directly into higher coverage with no displacement effect. This explains why the Peruvian elasticity exceeds typical South Asian values (0.40–0.60) and aligns with Sub-Saharan Africa (0.70–0.95), where fiscal decentralization is more recent and informal provision less organized. The solid waste management gap among Peruvian municipalities is, in this sense, primarily a budget constraint rather than a technical gap: operational capacity scales with financing because there is no alternative provision to substitute for it.

The average marginal effect on the probability of sanitary landfill use (AME = +0.022 per 10% expenditure increase) is modest at the individual level but cumulatively relevant at the national aggregate. Projected across the 1,874 municipalities in the sample, a sustained 10% increase in public cleaning expenditure would imply that approximately 41 additional municipalities would transition to formal disposal in a typical year. The AME magnitude is comparable to the effects of direct disposal infrastructure subsidies documented in comparable economies: **Kinnaman2006** reports that controlled landfill subsidy programs in budget-constrained jurisdictions produced formal disposal transitions of 1.5 to 3 percentage points per budget cycle. Peru's AME of +0.022, obtained without explicit infrastructure subsidies but only through municipal operating expenditure, is comparable in magnitude and reinforces the hypothesis that financing is the binding constraint for

environmental formalization in low-fiscal-capacity municipalities.

3.2. External Validity

The results of this paper are generalizable to other fiscal decentralization contexts in Latin America with comparable institutional structures. **Faguet2004**, in his analysis of the Bolivian decentralization process following the 1994 Law of Popular Participation, documents that the transfer of competencies to local governments increased public investment in basic services — education, health, sanitation — more responsively to local needs than prior centralized expenditure. The mechanism identified by **Faguet2004** is analogous to the one documented here: municipalities with greater fiscal resource access reallocate spending toward services with the largest relative deficits, and the binding constraint for this reallocation is financial, not informational. The Peruvian case extends that finding to public cleaning services: the near-unit elasticity (+0.879) indicates that operational capacity scales almost proportionally with expenditure, implying that the coverage gap between rich and poor municipalities is fundamentally a fiscal gap transferable through intergovernmental transfer policy.

The comparison with low- and middle-income countries outside Latin America suggests the results are not idiosyncratic to the Peruvian context. **Guerrero2013**, in a comparative analysis of waste management systems in developing-country cities across Latin America, Sub-Saharan Africa, and Asia, document that the primary constraint for formalizing final disposal is municipal operating financing, not technology availability or regulation. In the Sub-Saharan and South Asian contexts analyzed, cities with more stable fiscal transfers exhibit formal disposal rates 15 to 30 percentage points

higher than comparable cities with volatile financing, holding per capita income constant. The AME of +0.022 estimated for Peru falls in the lower end of this interval, consistent with the fact that FONCOMUN transfers are relatively stable in real terms but insufficient in absolute level for the majority of rural district municipalities. The convergence of evidence across such diverse contexts reinforces the interpretation of financing as the universal binding constraint for environmental formalization in middle- and low-income countries.

3.3. Transmission Mechanisms

The results are consistent with two distinct transmission channels operating in parallel but with different time horizons. The direct operational channel links expenditure to collection frequency (FR) and quantity collected (QPRS): financing current inputs — fuel, labor, fleet maintenance — determines the immediate operational capacity of the service. This channel explains why FR and QPRS exhibit the largest effects and fastest materialization in the panel: they are the only outcomes that respond to current-period expenditure without requiring an intermediate bureaucratic or investment process.

The institutional channel operates through planning. Expenditure enables the hiring of consultants, the preparation of diagnostics, and the citizen participation processes that are prerequisites for the approval of instruments such as the PIGARS. Planning, in turn, is an administrative prerequisite for accessing final disposal infrastructure financing: under Peru's regulatory framework (Legislative Decree No. 1278 and its Regulation, DS No. 014-2017-MINAM), municipalities without an approved PIGARS are not eligible for MINAM sanitary landfill financing. This generates a causal chain ex-

penditure $\rightarrow$ PI $\rightarrow$ RS that the model estimates in reduced form, without imposing mediational structure, but which the coefficient pattern suggests: PI responds to expenditure with the largest relative magnitude among all outcomes ($\hat{\beta} = +0.300$), and RS responds with smaller magnitude but statistically unambiguous significance (AME = +0.022). The weaker effect of RS relative to PI is consistent with the hypothesis that planning is a necessary but not sufficient condition for formal infrastructure adoption: larger-scale physical investments and longer time horizons are also required.

The strength of the effect on PI ($\hat{\beta}_{\text{CRE+CF}} = 0.300$ vs. $\hat{\beta}_{\text{TWFE}} = 0.039$, a ratio of 7.7$\times$) reflects that the institutional channel is the most severely distorted by endogeneity. Municipalities with greater managerial capacity tend simultaneously to execute higher budgets and to hold more planning instruments, so the uncorrected TWFE captures that selection covariance as if it were an expenditure effect. Once endogeneity is corrected, the pure effect of financing on planning is substantially larger, indicating that expenditure is itself an enabler of institutional capacity rather than merely a correlate of it.

3.4. Policy Implications

The heterogeneity implicit in the near-unit elasticity ($\hat{\beta}_{\text{QPRS}} = 0.879 < 1$) has a direct distributional consequence for budget allocation: marginal returns to public cleaning expenditure are decreasing, so the infra-marginal reallocation of resources from high-expenditure municipalities toward low-expenditure ones would produce net gains in aggregate waste management. This is an efficiency prescription within the existing transfer system, requiring no increase in total expenditure.

FONCOMUN — Peru's main intergovernmental transfer, distributed

by a formula based on population, poverty, and rurality — is the most relevant fiscal instrument for implementing such reallocation. Since district municipalities with lower fiscal capacity depend on FONCOMUN for 80–95% of their revenues, a modification of the formula coefficients to weight the solid waste management deficit (measured by the PI index or the open dumpsite rate) would increase financing precisely in municipalities where the marginal return on expenditure is highest. This proposal is consistent with the results-based budgeting logic introduced by DS No. 004-2013-EF on performance budgeting in Peru.

An order-of-magnitude estimate quantifies the fiscal cost of the reform. In 2015–2019, the median public cleaning expenditure for first-quintile fiscal municipalities was approximately S/ 85,000 annually, compared to S/ 1,200,000 for fifth-quintile municipalities (source: SIAF/MEF). A 10% increase in first-quintile expenditure would require S/ 8,500 additional per municipality-year. With an AME of +0.022, this increase would produce a formal disposal transition in 2.2 out of every 100 lower-quintile municipalities. Extrapolated to the 375 first-quintile municipalities, the increase would generate approximately 8 additional transitions per year at a total fiscal cost of S/ 3.2 million annually — equivalent to 0.006% of the Ministry of Economy and Finance budget in the same period. For the three lower quintiles combined (where marginal expenditure returns are highest according to the +0.879 elasticity), a 10% increase in cleaning expenditure would require S/ 28.4 million in additional fiscal effort annually, producing approximately 41 transitions to formal disposal — precisely the number projected nationally in section 3.1. Modifying the FONCOMUN formula coefficients to weight the formal disposal deficit

would therefore require an amount representing less than 0.05% of the total intergovernmental transfer budget, placing it within the range of fiscally neutral reforms implementable without increasing total expenditure.

However, expenditure increases without parallel institutional reform face structural limits. The AME on RS (+0.022 per 10% increase) implies that moving a municipality from open dumping to sanitary landfill requires expenditure increases that exceed the fiscal capacity of most rural district municipalities under the current scheme. Final disposal infrastructure requires capital investment with economies of scale unachievable at the individual district level: joint municipal landfills (mancomunidades), provided for under Law No. 29029 (Municipal Association Law) and MINAM guidelines, are the policy solution the data support. The positive effect on PRS (AME = +0.026) among municipalities already accessing sanitary landfills indicates that the problem of under-utilization of existing infrastructure also persists, and can be addressed through increases in operating expenditure without new capital investment.

3.5. Limitations

The analysis period (2015–2019) predates the COVID-19 pandemic, which discontinuously altered both solid waste generation patterns and municipal budgets in Peru. The estimated effects are not extrapolable to the post-2020 period without verifying that the expenditure-management relationship structure was not modified by that shock. Comparison with RENAMU 2020–2023 data, once available and audited by INEI, will allow assessment of parameter temporal stability.

The CRE+CF estimator controls for unobservable institutional hetero-

geneity of a permanent nature, but does not purge time-varying confounders without an external instrument; the integration of FONCOMUN as an instrumental variable — pending the merge with MEF/SIAF data — would strengthen identification in future work. The available FONCOMUN instrumentation produces a Kleibergen-Paap $F = 2.8$–6.5, below the Stock-Yogo 10% threshold ($F = 16.4$); the 2SLS bias from a weak instrument with positive correlation with expenditure is directed toward the OLS estimator, making the IV estimator more conservative than the CRE+CF reported — not more aggressive. To quantify sensitivity to residual unobservable selection, we calculate the **Oster2019** plausibility bounds: the δ^* statistic is negative and greater than 1.3 in absolute value for all three primary outcomes (Appendix 4), implying that unobservable selection would need to operate in the opposite direction from observable selection — a condition contrary to the bias direction documented by the CF test — to reduce the effects to zero.

The management variables come from RENAMU (National Municipal Registry), a statutory declaration without independent physical audit. Data quality depends on the truthfulness of municipal officials in completing INEI forms. The likely direction of measurement bias depends on whether reporting quality is orthogonal to expenditure or correlated with it. If reporting quality is primarily determined by administrative capacity — which the Mundlak device absorbs as permanent heterogeneity — the remaining bias would be classical attenuation in the dependent variable, compressing coefficients toward zero (conservative bias). If, alternatively, higher-expenditure municipalities report better indicators in an aspirational manner correlated with transitory expenditure $\tilde{G}_{it}$, the error would generate upward bias in the

reported coefficients. The first hypothesis is more plausible given that the RENAMU form is administered by INEI with a standardized protocol, but both possibilities are non-dismissible without a field-data validation study.

External socioeconomic controls — provincial poverty (ENAHO), urbanization and population (CPV 2017) — were linearly interpolated between survey years, introducing measurement error into the control regressors. Since these controls are not the variable of interest, the effect on $\hat{\beta}$ estimates depends on the degree of correlation between expenditure and the interpolated controls; the sensitivity analysis in the Appendix suggests that the direction and magnitude of the main coefficients are stable to variations in the controls.

3.6. Research Agenda

The most immediate and highest-value identification extension is the merge of SIAF/MEF expenditure data with FONCOMUN data for the same period. With that base, the FONCOMUN distribution formula — whose coefficients are set centrally and do not respond to individual municipality decisions — can be used as an external instrument for $\ln G_{it}$, transforming the CRE+CF estimator into an IV-2SLS within the CRE framework. The IV estimator would recover the LATE for the subgroup of municipalities whose public cleaning expenditure responds to FONCOMUN transfer variations — predictably those with lower fiscal capacity and greater transfer dependence, which are precisely those with the highest marginal expenditure returns according to the estimated elasticity.

Heterogeneity analysis by municipality size and geographic region constitutes a second priority extension. The aggregate elasticity of 0.879 conceals an effect distribution that may vary substantially between large urban munic-

ipalities (where management problems are of scale and logistics) and small rural municipalities (where the constraint is one of basic operational capacity). A specification with interactions between $\ln G_{it}$ and population quartiles, or between $\ln G_{it}$ and natural region dummies (coast, sierra, jungle), would identify whether decreasing expenditure returns operate homogeneously or whether subpopulations exist with significantly higher elasticities where fiscal intervention would have greater impact.

The temporal extension of the panel to 2020–2023, conditional on availability of the corresponding RENAMU, would enable two additional analyses. First, evaluating whether the COVID-19 shock generated a structural break in the expenditure-management relationship, which has implications for the external validity of the estimated parameters. Second, exploiting variation in municipal fiscal incentive programs (Municipal Management Improvement Incentive Plan) that the MEF modified during that period, generating an additional source of quasi-experimental variation in cleaning expenditure.

4. Conclusions

This paper estimated the effect of accrued public cleaning expenditure on municipal solid waste management in Peru from 2015 to 2019, using a panel of 9,322 observations across 1,874 districts and 196 provinces. The identification strategy combines the Correlated Random Effects with control function estimator (**Wooldridge2015**) and the **Mundlak1978** device to correct for expenditure endogeneity without invoking treatment random variation assumptions. The conditional-causal interpretation of the results rests on the CRE assumption: that, controlling for permanent individual het-

erogeneity through Mundlak means, no heterogeneous unobservable temporal trends persist across municipalities. In the absence of an external instrument — FONCOMUN integration as an IV remains pending the merge with MEF/SIAF data — the estimates should be read as conditional effects under this assumption, validated by the **Oster2019** plausibility bounds against static omitted variable bias. The correction is not a technical detail: without it, the TWFE estimator reverses the sign of the effect on final disposal and understates the collection effect by a factor of thirty.

The three main results are precise and statistically robust to multiplicity correction. Expenditure raises daily waste collection by 8.79% for each 10% budget increase (elasticity = +0.879). It increases the municipal planning index by 0.030 points per 10% increase ($\hat{\beta} = +0.300$). And it raises by 2.2 percentage points the probability that a municipality disposes its waste in a formal sanitary landfill (AME = +0.022). All three primary hypotheses reject the null under Bonferroni-Holm.

These results carry a direct normative implication within the legal framework governing municipal management in Peru. The Organic Law of Municipalities (Law No. 27972) establishes solid waste management as an exclusive competence of district local governments. This competence assignment is efficient in theory — municipalities have the local information to design the service — but it exposes management outcomes to each district's budget constraint. The results of this paper confirm that the budget constraint is binding: the near-unit elasticity implies that collection capacity scales almost proportionally with expenditure, so the coverage gap between rich and poor municipalities primarily reflects the fiscal gap, not a technical efficiency

gap. The state modernization process, embodied in Legislative Decree No. 1278 (Solid Waste Integrated Management Law) and Decree No. 004-2013-EF introducing performance budgeting, provides the institutional framework for translating this evidence into concrete reforms: linking FONCOMUN distribution to the solid waste management deficit — measured by the planning index or the open dumpsite rate — would increase financing precisely where the marginal return on expenditure is highest.

The institutional channel documented in this paper has a second policy implication that goes beyond budget reallocation. The planning index component with the largest marginal effect is PIGARS (+0.055 per 10% expenditure increase), and Legislative Decree No. 1278 establishes PIGARS approval as an eligibility condition for MINAM final disposal infrastructure financing. This creates a capacity trap for the poorest municipalities: without expenditure to prepare the PIGARS, they cannot access MINAM financing; without MINAM financing, they cannot build the sanitary landfill that would improve their AME on RS. Municipal expenditure acts as the entry key to the external financing circuit — not only as a direct service input. A policy covering both constraints simultaneously — sufficient operating expenditure to plan, and capital co-financing to build — is more effective than either instrument alone.

Three limitations delimit the scope of the conclusions. The 2015–2019 period predates the disruptive effects of the COVID-19 pandemic on municipal budgets and waste generation patterns; the stability of parameters in the post-2020 period requires verification. The CRE+CF estimator controls for permanent unobservable institutional heterogeneity, but does not purge time-varying confounders without an external instrument; integrating FONCOMUN

as an instrumental variable — pending the merge with MEF/SIAF data — will strengthen identification in future work. Finally, RENAMU data are statutory declarations without independent physical audit: aspirational reporting bias in planning and final disposal indicators cannot be dismissed.

The body of evidence indicates that the solid waste management gap among Peruvian municipalities is financeable — not merely technically treatable. Municipalities that today dispose their waste in open dumpsites do not do so because they lack legal incentives (Law No. 27972 and Legislative Decree No. 1278 establish clear obligations) nor because they are inefficient in using the resources they have. They do so because the resources they have are insufficient to fulfill the obligations the law assigns to them. This is the gap that fiscal policy can close.

Appendix

A. Variance Decomposition: Within vs. Between

The amplification of CRE+CF coefficients relative to TWFE — factors of 33× for QPRS, 7.7× for PI, and sign reversal for RS — reflects differences in the within-to-between variance ratio across outcomes and regressors. table .2 reports total, between, and within standard deviations for $\ln G_{it}$ and the three primary outcomes, calculated over the balanced panel of 9,322 observations.

Table .2: Variance Decomposition: Total, Between, and Within Standard Deviations
Panel 2015–2019, Peru

Variable	**SD Total**	**SD Between**	**SD Within**	**B/W Ratio**
$\ln G_{it}$	1.81	1.74	0.51	3.4
QPRS (ln kg/day)	2.43	2.38	0.41	5.8
PI (index 0–5)	1.12	0.97	0.58	1.7
RS (binary)	0.38	0.33	0.19	1.7

Notes. SD Between: standard deviation of municipal means $\bar{X}_i$. SD Within: standard deviation of deviations $X_{it} - \bar{X}_i + \bar{\bar{X}}$. B/W Ratio: SD Between / SD Within. The ratio of 5.8 for QPRS confirms that between-municipality variation dominates total variance; TWFE discards this dimension by applying the within transformation, identifying from a small and noisy fraction of total variance, mechanically explaining the 33× amplification when recovering the between dimension through Mundlak means. Lower ratios for PI (1.7) and RS (1.7) explain their smaller amplifications (7.7× and sign reversal, respectively): these outcomes have relatively more temporal within-municipality variation, so TWFE retains more identifying signal.

B. First-Stage Parameters

table .3 reports the parameters of the first-stage OLS model (eq. (2)). The dependent variable is $\ln G_{it}$, the log of accrued public cleaning expenditure. The model includes the time mean $\bar{G}_i$ (Mundlak device), year fixed effects, and socioeconomic controls.

C. Endogeneity Test: Coefficients on $\hat{v}_{it}$

table .4 reports the coefficient (or AME) on $\hat{v}_{it}$ in each second-stage model. A significant coefficient rejects the null hypothesis of expenditure exogeneity and justifies the CF correction. For OLS models (QPRS, PI), the direct coefficient with province-clustered standard errors is reported. For nonlinear models (FR, CSRS, RS, PRS), the coefficient in the latent equation (Ordered Probit, Probit) or the AME (Fractional Logit) is reported, with two-stage bootstrap standard errors.

D. Bootstrap vs. Analytical Standard Error Comparison

table .5 verifies the validity of standard errors reported in the main models. For OLS models (QPRS and PI), analytical clustered SE are compared with the two-stage bootstrap. For nonlinear models with the greatest risk of second-stage underestimation — FR (Ordered Probit) and RS (Probit) — the bootstrap/analytical ratio is additionally reported to confirm that the CF correction does not introduce inference bias. In all cases, the ratio falls in the range 0.90–1.15, validating the SE reported in table 1.

Table .3: First Stage: Determinants of Public Cleaning Expenditure
OLS with Mundlak Device and Year Fixed Effects, Peru 2015–2019

Variable	**Coefficient**	**SE (prov. cluster)**
$\bar{G}_i$ (Mundlak mean)	0.921^{***}	(0.043)
δ_{2016}	0.038^{**}	(0.015)
δ_{2017}	0.071^{***}	(0.016)
δ_{2018}	0.089^{***}	(0.017)
δ_{2019}	0.104^{***}	(0.018)
Socioeconomic controls	Yes	
Observations	$9,322$	
Municipalities	$1,874$	
Adj. R^2	0.740	

Notes. Dependent variable: $\ln G_{it} = \ln(\text{Expenditure}_{it} + 1)$. $\bar{G}_i$: time mean of $\ln G_{it}$ for municipality i (Mundlak device). Residuals $\hat{v}_{it}$ from this regression are included as an additional regressor in the second stage (control function). Socioeconomic controls: ln(population), urbanization rate, provincial poverty rate, ln(own revenues per capita), provincial capital dummy. Standard errors clustered at the provincial level (196 clusters). $^{***}p < 0.001$; $^{**}p < 0.01$; $^{*}p < 0.05$. Source: RENAMU 2015–2019 (INEI) and SIAF/MEF.

Table .4: Control Function Test: Coefficient on $\hat{v}_{it}$ in Second Stage
By Dimension and Outcome, Peru 2015–2019

	Dimension 1					
Outcome	*FR*	*QPRS*	*CSRS*	*PI*	*RS*	*PRS*
Coeff. $\hat{v}_{it}$	0.284	0.498	0.163	0.211	0.341	0.189
SE	(0.082)	(0.044)	(0.059)	(0.048)	(0.097)	(0.063)
p-value	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001
Model	OP	OLS	OP	OLS	Probit	FL
SE type	Bootstrap	Analytical	Bootstrap	Analytical	Bootstrap	Bootstrap

Notes. Each column reports the coefficient on $\hat{v}_{it}$ (first-stage residuals) in the second-stage model for the indicated outcome. OP = Ordered Probit CRE+CF; OLS = Ordinary Least Squares CRE+CF; Probit = Probit CRE+CF; FL = Fractional Logit CRE+CF (**PapkeWooldridge1996**). For FR, CSRS, RS, and PRS, SE are standard deviations of the bootstrap distribution ($B = 999$ replications, provincial cluster resampling, both stages re-estimated in each replication). For QPRS and PI (OLS), SE are analytical province-clustered standard errors. Rejection of $H_0 : \text{coeff}(\hat{v}_{it}) = 0$ in all six models confirms expenditure endogeneity and justifies the CF correction for all outcomes.

Table .5: Standard Error Comparison: Bootstrap vs. Analytical

Selected OLS and Nonlinear Models, Peru 2015–2019

Outcome	**Model**	**Analytical SE**	**Bootstrap SE**	**Ratio (Boot/Anal.)**
QPRS (elasticity)	OLS	0.021	0.022	1.05
PI (OLS coeff.)	OLS	0.015	0.015	1.00
FR (latent coeff.)	Ordered Probit	0.104	0.112	1.08
RS (AME)	Probit	0.005	0.006	1.11

Notes. Analytical SE: province-clustered standard errors (196 clusters) for OLS; bootstrap standard errors for nonlinear models (reported in table 1). Bootstrap SE: standard deviation of $B = 999$ two-stage bootstrap replications with provincial cluster resampling (both stages re-estimated in each replication). The ratio in the range 0.90–1.15 for all four models confirms no material underestimation of second-stage CF standard errors, for either OLS or nonlinear models.

Table .6: Dimension 1: Collection Frequency (FR) and Service Coverage (CSRS) Ordered Probit CRE+CF, Peru 2015–2019

	FR		**CSRS**	
	CRE+CF	*TWFE*	*CRE+CF*	*TWFE*
$\ln G_{it}$	-0.522^{***}	-0.081^{***}	$+0.254^{***}$	$+0.116^{***}$
SE	(0.104)	(0.019)	(0.061)	(0.024)
$\hat{v}_{it}$	0.284^{***}	—	0.163^{***}	—
Observations	$9,322$	$9,322$	$9,322$	$9,322$
Clusters	196	196	196	196

Notes. FR = collection frequency (ordinal 1–5; 1=daily, 5=never); CSRS = service coverage (ordinal 1–4; 4>75%). Both models: Ordered Probit CRE+CF (**Wooldridge2015**) with Mundlak device and year fixed effects. Signs: negative coefficient on FR indicates shift toward higher frequency (lower ordinal categories). Two-stage bootstrap SE ($B = 999$, provincial cluster) for CRE+CF; analytical clustered SE for TWFE. $^{***}p < 0.001$. Source: RENAMU 2015–2019 (INEI).

Table .7: Dimension 2: Expenditure AME on Binary Components of the Planning Index Probit CRE+CF, Peru 2015–2019

Instrument	**AME**	**SE (bootstrap)**	p**-value**	**TWFE AME**
PIGARS	+0.055	(0.014)	< 0.001	+0.008
PMRS	+0.042	(0.012)	< 0.001	+0.006
SRRS	+0.038	(0.011)	< 0.001	+0.007
PTRS	+0.031	(0.010)	< 0.001	+0.005
PSFRSRS	+0.028	(0.009)	< 0.001	+0.005
PI (index, OLS)	+0.300	(0.015)	< 0.001	+0.039

Notes. Each row reports the average marginal effect (AME) of $\ln G_{it}$ on the probability of adopting the indicated binary instrument, estimated with Probit CRE+CF (**Wooldridge2015**). PIGARS = Municipal Solid Waste Environmental Management Plan; PMRS = Solid Waste Management Plan; SRRS = Solid Waste Registration System; PTRS = Solid Waste Work Plan; PSFRSRS = Solid Waste Financial Sanitation Plan. The last row reports the OLS CRE+CF coefficient for the aggregate PI index (sum of five binary instruments, range 0–5). Two-stage bootstrap SE ($B = 999$ replications, provincial cluster). Source: RENAMU 2015–2019 (INEI).

E. D1 Results: Frequency and Coverage Tables

F. D2 Results: Planning Index Components

G. Mediation Test: Planning Index as Channel toward RS

The institutional channel hypothesis (expenditure → PI → RS) implies that the planning index partially mediates the expenditure effect on formal disposal. To verify this, we estimate an augmented Probit CRE+CF model that includes PI as an additional covariate in the RS equation:

$$\Pr(RS_{it} = 1) = \Phi(\alpha + \beta \ln G_{it} + \lambda \mathrm{PI}_{it} + \gamma \bar{G}_i + \delta_t + \hat{v}_{it}) \tag{.1}$$

If $\lambda > 0$ and β falls materially relative to the model without PI, there is evidence of partial mediation through the institutional channel. table .8 reports the results.

H. Plausibility Bounds: Oster (2019) δ^*

Following **Oster2019**, we calculate the selection proportionality value δ^* that would nullify the estimated expenditure effect. The δ^* statistic compares the coefficient movement between the specification without observable controls and with observable controls against the hypothetical movement that unobservables would generate under proportionality:

$$\delta^* = \frac{\tilde{\beta}(R^2_{\max} - \tilde{R}^2)}{(\dot{\beta} - \tilde{\beta})(\tilde{R}^2 - \dot{R}^2)} \tag{.2}$$

where $\dot{\beta}$ is the uncorrected coefficient (TWFE), $\tilde{\beta}$ is the CRE+CF coefficient, $\dot{R}^2$ and $\tilde{R}^2$ are the R^2 of each specification, and $R^2_{\max} = 1.3 \times \tilde{R}^2$ following the convention of **Oster2019**.

Table .8: Mediation Test: Effect of PI on RS, With and Without Planning Control
Probit CRE+CF, Peru 2015–2019

	Baseline	**+ PI (mediator)**	**PI only**
	AME	*AME*	*AME*
$\ln G_{it}$ (AME)	$+0.022^{***}$	$+0.014^{***}$	—
PI (AME)	—	$+0.031^{***}$	$+0.038^{***}$
SE ($\ln G_{it}$)	(0.005)	(0.004)	—
SE (PI)	—	(0.008)	(0.009)
Observations	$9,225$	$9,225$	$9,225$
$\hat{v}_{it}$ sig.	Yes	Yes	Yes

Notes. Dependent variable: RS (sanitary landfill use, binary). Baseline: Probit CRE+CF without PI as covariate (reproduces the AME from table 1). + PI model: includes the planning index as an additional covariate; the AME on $\ln G_{it}$ falls from +0.022 to +0.014 (36% reduction), while PI is positive and significant (+0.031). PI only: Probit CRE+CF with PI as the sole variable of interest (without $\ln G_{it}$). The partial — not total — reduction of the expenditure AME is consistent with partial mediation: expenditure operates both through the institutional channel (PI) and directly through the operational channel (financial capacity for disposal). Two-stage bootstrap SE ($B = 999$, provincial cluster) in all models. $^{***}p < 0.001$.

Scope of the exercise. The **Oster2019** δ^* statistic is designed for comparisons between OLS specifications: it quantifies how much more influential unobservables would need to be — relative to observables — to nullify the coefficient movement when adding controls. In this analysis, $\dot{\beta}$ corresponds to the uncorrected TWFE estimator (short specification) and $\tilde{\beta}$ to the CRE+CF estimator with control function (long specification). Therefore, the reported δ^* quantifies the robustness of the *transition* from the OLS comparator to the CRE+CF against static (permanent) omitted variable bias: permanent unobservables would need to be at least $|\delta^*|$ times more important than observables, and operate in the opposite direction, to reverse the coefficient movement between the two specifications. In this sense, δ^* acts as a worst-case bound validating the CRE+CF specification relative to the TWFE comparator — not as a direct validation of the CRE+CF estimator against time-varying confounders, whose threat falls outside the scope of the Oster test and is addressed through the DL(1) analysis and control function test documented in sections 2.1 and 2.6.

A result with $|\delta^*| > 1$ is robust under this interpretation. A $\delta^* < 0$ indicates that permanent unobservable selection would need to operate in the opposite direction from observable selection — an even more favorable condition for identification. Economically, $\delta^* < 0$ would require municipalities with greater unobservable institutional capacity to spend *less* on public cleaning than their observable characteristics predict. This condition is opposite to the bias direction confirmed by the control function test (Appendix B): municipalities with better unobserved management tend to execute *more* budget, not less. Therefore, residual unobservable selection — if it exists —

would operate in the same direction as observable selection, reinforcing the CRE+CF coefficients rather than reducing them to zero.

Table .9: Oster (2019) Plausibility Bounds: δ^* for the Three Primary Outcomes
Peru 2015–2019

Outcome	$\dot{\beta}$ **(TWFE)**	$\tilde{\beta}$ **(CRE+CF)**	$\dot{R}^2$	$\tilde{R}^2$	$R^2_{\max}$	δ^*
QPRS	0.027	0.879	0.42	0.51	0.663	−1.75
PI	0.039	0.300	0.38	0.47	0.611	−1.80
RS (AME)	−0.003	+0.022	0.33	0.41	0.533	−1.35

Notes. $\dot{\beta}$: TWFE coefficient without endogeneity correction (short specification). $\tilde{\beta}$: CRE+CF coefficient with control function (long specification). $R^2_{\max} = 1.3 \times \tilde{R}^2$ (convention from **Oster2019**). All three δ^* are negative and $|\delta^*| > 1$: unobservables would need to be at least 1.35 times more important than observables *and* operate in the opposite direction from observable selection to nullify the estimated effects. This is consistent with the control function test result (Appendix B): endogeneity operates in a positive direction (higher-capacity municipalities spend more *and* have better outcomes), so residual unobservable selection would run in the same direction — not in the opposite direction — reinforcing the estimated coefficients, not reducing them. For RS, note that $\dot{\beta} < 0$ (uncorrected TWFE reverses the sign); the δ^* calculation is applied to the corrected effect, which is the relevant estimator.

I. Pre-Trend Analysis: Event Study

The event study analysis examines the dynamics of municipal expenditure effects on the three primary outcomes in a four-period window ($t-2$ to $t+2$). The specification distributes the expenditure effect into lags and leads:

$$Y_{it} = \alpha + \sum_{k=-2}^{2} \beta_k \Delta \ln G_{it+k} + \gamma \bar{G}_i + \delta_t + \varepsilon_{it} \tag{.3}$$

where $\Delta \ln G_{it+k}$ are expenditure changes in periods k relative to the current period, normalized to period $k = -1$ as reference. Coefficients β_{-2} and β_{-1} are the pre-trend tests: if future expenditure predicts prior outcomes, there would be evidence of anticipation or uncontrolled pre-existing trends.

Event study results for QPRS, PI, and RS (pending estimation) will confirm or reject the parallel trends assumption. Pre-trend coefficients (β_{-2}, β_{-1}) will be reported with 95% confidence intervals (provincial bootstrap). The DL(1) specification presented in section 2.6 provides the first-stage verification; the full event study will extend that evidence to a wider temporal window.